

Weight of ASTM A325 or A490 high strength bolts

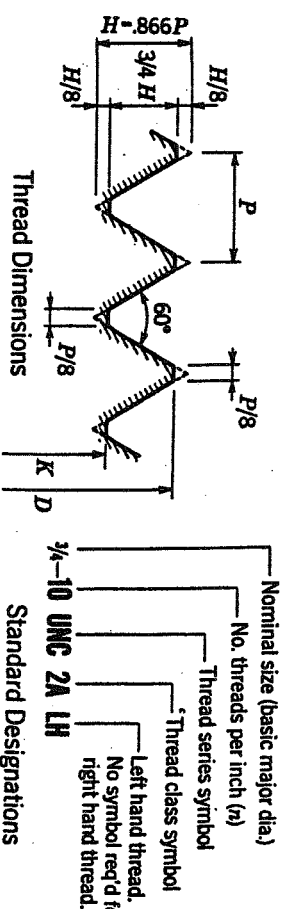
Heavy hex structural bolts with heavy hex nuts in pounds per 100

Shulman

SCREW THREADS

Unified Standard Series—UNC and 4 UN

ANSI B1.1—1960



Thread Dimensions

Standard Designations

Basic Major Diameter D	Root K	Area			Threads per inch n	Basic Major Diameter D	Root K	Area			Threads per inch n
		Gross A_D	Root A_R	Tensile Stress				Gross A_D	Root A_R	Tensile Stress	
1/4	.185	.049	.027	.032	20	2 1/4	2.425	5.940	4.62	4.93	4
5/16	.294	.110	.068	.078	16	3	2.675	7.069	5.62	5.97	4
3/8	.400	.195	.126	.142	13	3 1/4	2.925	8.296	6.72	7.10	4
1/2	.507	.307	.202	.226	11	3 1/2	3.175	9.621	7.92	8.33	4
5/8	.620	.442	.302	.334	10	3 3/4	3.425	11.045	9.21	9.65	4
3/4	.731	.601	.419	.462	9	4	3.675	12.566	10.6	11.1	4
7/8	.838	.785	.551	.606	8	4 1/4	3.925	14.186	12.1	12.6	4
1 1/8	.994	.994	.693	.763	7	4 1/2	4.175	15.904	13.7	14.2	4
1 1/4	1.064	1.227	.890	.969	7	4 3/4	4.425	17.721	15.4	16.0	4
1 3/8	1.158	1.485	1.05	1.16	6	5	4.675	19.635	17.2	17.8	4
1 1/2	1.283	1.767	1.29	1.41	6	5 1/4	4.925	21.648	19.1	19.7	4
1 3/4	1.490	2.405	1.74	1.90	5	5 1/2	5.175	23.788	21.0	21.7	4
2	1.711	3.142	2.30	2.50	4 1/2	5 3/4	5.425	25.967	23.1	23.8	4
2 1/4	1.961	3.976	3.02	3.25	4	6	5.675	28.274	25.3	26.0	4
2 1/2	2.175	4.909	3.72	4.00	4						

$$a \text{ Tensile stress area} = 0.7854 \left(D - \frac{.6743}{n} \right)^2$$

^b For basic major diameters of 1/4 to 4 in. incl., thread series is UNC (coarse); for 4 1/4 in. diameter and larger, thread series is UNF.
^c 2A denotes Class 2A fit applicable to external threads, 2B denotes corresponding Class 2B fit for internal threads.

MINIMUM LENGTH OF THREAD ON BOLTS

ANSI B18.2.1—1965

Length of Bolt	Diameter of Bolt, D , inches											
	1/4	3/8	1/2	5/8	3/4	7/8	1	1 1/4	1 1/2	1 3/4	2	2 1/2
To 6 in. incl.	3/4	1	1 1/4	1 1/2	1 3/4	2	2 1/4	2 1/2	2 3/4	3	3 1/4	3 1/2
Over 6 in.	1	1 1/4	1 1/2	1 3/4	2	2 1/4	2 1/2	2 3/4	3	3 1/4	3 1/2	3 3/4

Note 1. Thread length for bolts up to 6 in. long is $2D + 1/4$. For bolts over 6 in. long, thread length is $2D + 1/2$. These proportions may be used to compute thread length for diameters not shown in the table. Bolts which are too short for listed or computed thread lengths are threaded as close to the head as possible.
 Note 2. For thread lengths for high strength bolts, refer to "Specifications for Structural Joints Using ASTM A325 or A490 Bolts," in Part 5 of this manual.

Length Under Head inches	Diameter of Bolt in inches									
	1/2	3/8	1/4	7/8	1	1 1/8	1 1/4	1 3/8	1 1/2	1 3/4
1 1/4	16.5	29.4	47.0	74.4	104	148	197	261	333	344
1 1/2	17.8	31.1	49.6	77.0	109	154	203	267	339	350
1 3/4	19.2	33.1	52.2	81.9	114	160	210	274	349	360
2	20.5	35.3	55.3	86.1	119	167	217	281	355	366
2 1/4	21.9	37.4	58.4	90.3	124	174	224	288	362	373
2 1/2	23.3	39.8	61.6	94.6	130	181	231	295	369	380
2 3/4	26.1	43.9	67.8	98.8	135	188	238	302	376	387
3	27.4	46.1	70.9	103	141	195	245	309	383	394
3 1/4	28.8	48.2	74.0	107	146	202	252	316	390	401
3 1/2	30.2	50.4	77.1	111	151	209	259	323	397	408
3 3/4	31.6	52.5	80.2	116	157	216	266	330	404	415
4	33.0	54.7	83.3	120	162	223	273	337	411	422
4 1/4	34.3	56.9	86.4	124	168	230	280	344	418	429
4 1/2	35.7	59.0	89.5	128	173	237	287	351	425	436
4 3/4	37.1	61.2	92.7	133	179	244	294	358	432	443
5	38.5	63.3	95.8	137	184	251	301	365	439	450
5 1/4	39.9	65.5	98.9	141	190	258	308	372	446	457
5 1/2	41.2	67.7	102	146	196	265	315	379	453	464
5 3/4	42.6	69.8	105	150	201	272	322	386	460	471
6	44.0	71.9	108	154	207	279	329	393	467	478
6 1/4	45.4	74.1	111	158	212	286	336	400	474	485
6 1/2	46.8	76.3	114	163	218	293	343	407	481	492
6 3/4	48.2	78.5	118	167	223	299	350	414	488	499
7	49.6	80.6	121	171	229	306	357	421	495	506
7 1/4	51.0	82.8	124	175	234	313	364	428	502	513
7 1/2	52.4	84.9	127	179	240	320	371	435	509	520
7 3/4	53.8	87.1	130	183	246	327	378	442	516	527
8	55.2	89.2	133	187	251	334	385	449	523	534
8 1/4	56.6	91.4	137	191	257	341	392	456	530	541
8 1/2	58.0	93.6	140	195	262	348	400	463	537	548
8 3/4	59.4	95.8	144	199	268	355	407	470	544	555
9	60.8	98.0	148	203	273	362	414	477	551	562
Per inch additional and	5.5	8.6	12.4	16.9	22.1	28.0	34.4	42.5	49.7	56.8
For each 100 plain round washers add	2.1	3.6	4.8	7.0	9.4	11.3	13.8	16.8	20.0	23.1
For each 100 beveled square washers add	23.1	22.4	21.0	20.2	19.2	34.0	31.6	...	...	...

Note: This table conforms to weight standards adopted by the Industrial Fasteners Institute, 1965, updated for washer weights.